

ASAD IMRAN

Electronics & Embedded Systems Engineer

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PROFESSIONAL OVERVIEW

Electronics engineer with experience in embedded systems, hardware design, FPGA development, and high-speed digital electronics. Experienced in taking products from concept through design, prototyping, validation, and production. Comfortable working across hardware, firmware, debugging, system integration, and cross-functional engineering teams to deliver reliable embedded solutions

WORK HISTORY

RapidAnalysis | Sydney, Australia

Hardware Design Engineer | Feb 2025 to Aug 2026

- Sole designer of the 6-layer CM4/CM5 and 8-layer Jetson Orin Nano carrier boards, from requirements and reference designs through schematic capture, KiCad PCB layout, component selection within size and cost limits, and prototype release.
- CM4/CM5 carrier (120 × 40 mm): NVMe, Gigabit Ethernet, PoE (25 W module) and USB-C PD. Traced an NVMe link failure on the CM4 to module control signals and fixed it in Rev 2 ([Rev 1](#), [Rev 2](#)).
- Jetson Orin Nano carrier: 260-pin SO-DIMM breakout, M.2 NVMe, OCUlink and a 5 V rail designed for about 8 A. Released Rev A for PCBA.
- Ran ERC/DRC, design reviews, bring-up and debugging. Managed BOM optimisation, sourcing, supplier liaison, PCBA ordering and manufacturing support, including DFM/DFA.

Engineering Intern | Aug 2024 to Jan 2025

- Integrated an SPI-based e-ink display into an existing embedded platform and designed CM4 carrier boards from reference designs.

CARE, Pvt. Ltd. | Islamabad, Pakistan

Design Engineer | Apr 2022 to Jun 2023

- Developed and maintained backend components of a Lawful Interception System across legacy and replacement systems.
- Implemented secure storage, encryption and disaster recovery mechanisms and supported day-to-day operation and reliability.
- Built software simulators and stress test tools that generate workloads and measure performance and failure limits.

PROJECTS

RF Fingerprinting on RFSoc | Master's thesis, Macquarie University

- CNN classifier for a 12-device Zigbee dataset, quantised with HLS4ML and deployed on a Xilinx ZCU111. About 99% accuracy in Python and HLS4ML, 97.55% on the deployed 24-bit model, 3 to 4 ms per inference.

Multi-UART Data Acquisition | Final year project, NUST

- Wrote the FreeRTOS firmware for a team data acquisition system on an STM32F429, receiving four 460800-baud UART streams through DMA and double buffering with no data loss.

EDUCATION

Master of Engineering in Electronics Engineering

Macquarie University, Sydney
2023 to 2025

Bachelor of Electrical Engineering

NUST (SEECs), Islamabad
2018 to 2022

SKILLS

- Embedded & Sensing:** STM32, ARM Cortex-M, Raspberry Pi CM4/CM5, FreeRTOS, ADC, DMA, Real-Time Data Acquisition, Sensor Interfacing
- Communication Interfaces:** Gigabit Ethernet, PoE, USB-C PD, PCIe, UART, I²C, I²S, SPI
- Programming:** C, C++, C#, Java, Python, MATLAB
- Signal Processing & FPGA:** DSP, SDR, RF Signal Processing, Xilinx RFSoc, VHDL, HLS, HLS4ML
- Hardware Engineering:** PCB Design, Schematic Capture, KiCad, Signal Integrity
- Testing & Integration:** Oscilloscopes, Logic Analysers, Hardware Bring-up, Prototype Validation, Root Cause Analysis, System Integration, DFM/DFA, BOM and Sourcing